



SMART PLANT MONITORING

User Manual

MODEL GP-1

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Welcome to GrowthPulse

Thank you for choosing GrowthPulse, the smart plant monitoring system that watches over your plants around the clock and tells you exactly what they need. This manual covers everything: setting up your device, connecting it to your home Wi-Fi, pairing it with your GrowthPulse account, and getting the most out of every feature in the app.

If you only read one section, read **Quick Start**. It gets you from box to live readings in about five minutes.

The app lives at growthpulsecloud.com. It works in any browser on your phone, tablet, or computer, and it can be installed on your phone's home screen like a regular app (see "Install the app on your iPhone").

Safety and Care

- The GrowthPulse unit is powered by USB-C at 5 volts, the same kind of charger as most modern phones. Use any standard USB-C power adapter.
- The main electronics enclosure is **not waterproof**. Keep it out of rain and away from direct watering. Only the soil probes are designed to touch soil and water.
- The soil temperature probe (the metal-tipped probe on a cable) is waterproof and safe to bury in soil.
- The soil moisture probe may be inserted into soil up to its marked line. Do not submerge the electronics at the top of the probe.
- Operating temperature: the unit is happiest between 32°F and 120°F (0°C to 49°C). For outdoor use, shelter it from direct rain and harsh sun.
- There are no user-serviceable parts inside the enclosure.

In the Box

- 1x GrowthPulse smart monitoring unit (with built-in display)
- 1x waterproof soil temperature probe
- 1x air temperature and humidity sensor
- 1x capacitive soil moisture probe
- 1x USB-C power cable
- Quick start card with your unit's pairing code

Your unit's **pairing code** is also shown directly on the unit's built-in screen whenever it is powered, so you can never lose it.

Meet Your Device

The screen. The built-in display walks you through setup and then shows your unit's pairing code in large letters, along with a small status indicator: **ON** means your unit is connected and streaming, two dots (..) mean it is still connecting.

The PRG button. One button does everything:

- **Hold for 3 seconds** while the unit is running: forgets the saved Wi-Fi and reopens setup mode. Use this when you move to a new home, change your Wi-Fi password, or set the unit up somewhere new.
- You never need this button for day-to-day use.

The USB-C port. Power in. Plug it into any standard USB-C wall adapter.

The sensors. Three sensors feed your dashboard: air temperature and humidity (the vented sensor), soil temperature (the metal probe), and soil moisture (the flat blade probe). Place the soil probes in the pot or bed you want to monitor, near the plant's root zone but not touching the stem.

Quick Start

1. **Plug in your GrowthPulse** with the USB-C cable. The screen lights up and says it is starting.
2. **On your phone, open Wi-Fi settings** and join the network named **GrowthPulse-XXXXXX** (the exact name shows on the unit's screen).
3. A **setup page opens automatically**. Pick your home Wi-Fi from the list, type your Wi-Fi password, and tap save. If the page does not pop up within 15 seconds, open your browser and go to **192.168.4.1**.
4. The unit connects and its screen shows your **pairing code** in big letters with the ON indicator.
5. **Go to growthpulsecloud.com**, create your account, tap **Connect a device**, and enter the pairing code from the screen. Done, your plant is live.

Connecting to Wi-Fi (Detailed)

Your GrowthPulse connects to the internet through your home Wi-Fi. It only needs to do this once; afterward it reconnects on its own, even after power outages.

Setup mode. A brand-new unit (or one that has been reset) automatically opens setup mode: it broadcasts its own temporary Wi-Fi network named GrowthPulse-XXXXXX and shows those instructions on its screen.

Joining from your phone:

1. Open your phone's Wi-Fi settings and join the GrowthPulse-XXXXXX network. It has no password.
2. Your phone will detect that this network needs a sign-in and open the GrowthPulse setup page on its own.
3. If your phone says "this network has no internet access," choose **Keep trying Wi-Fi** or **Use this network anyway**. That is expected, the unit is not on the internet yet, that is why you are setting it up.
4. If the setup page never appears, open any browser and type **192.168.4.1** in the address bar. The setup page is there.

On the setup page: you will see the GrowthPulse welcome screen. Tap Configure WiFi, choose your home network from the list, enter your Wi-Fi password, and save.

Good to know:

- GrowthPulse uses **2.4 GHz Wi-Fi**, which has much better range through walls than 5 GHz. If your router broadcasts separate names for 2.4 and 5 GHz, pick the 2.4 GHz one.
- Setup mode stays open for 10 minutes. If it times out, the unit restarts and opens setup mode again.
- Turning off cellular data during setup helps some phones show the setup page faster.

Changing Wi-Fi later (new router, new password, new home): hold the **PRG button for 3 seconds**. The screen shows "Clearing Wi-Fi," the unit restarts into setup mode, and you repeat the steps above. Your account, plant data, and settings are not affected.

Your Account and Pairing

Creating an account. At growthpulsecloud.com choose Create an account. Enter your first and last name, email, and a password (entered twice to catch typos), and tell us how you grow: Hobbyist, Home grower, Farmer, or Commercial. This helps us tailor the experience.

Trying before buying. The **Explore the demo** button on the sign-in screen opens a fully working sandbox with sample plants, no account needed. Demo data is clearly separate and never mixes with a real account.

Pairing your device. From a new account you will see the welcome screen, tap **Connect a device**. You will be asked for:

- **How it connects:** Wi-Fi (most units) or through a LoRaWAN gateway (Farm Kit).
- **Plant name:** whatever you like, "Kitchen Basil," "Greenhouse Tomatoes."
- **The plant's home location:** a city or ZIP code. This matters! Weather and rain alerts follow the plant's home, not your phone, so a vacation on the other side of the world never confuses your rain forecast.
- **The pairing code:** the large code on your unit's screen.

A code that does not match a real GrowthPulse unit is rejected, so a typo cannot create a ghost device.

Privacy: every account's plants, history, journals, and alarms are private to that account.

The Live Dashboard

The Live tab is your plant's homepage.

The big three numbers across the top: air temperature, humidity, and soil moisture. Each is colored by status, green is ideal for your plant, amber is getting outside the comfort zone, red needs attention. A sensor that is not plugged in shows a dash and says "not connected" rather than pretending.

The moisture gauge and health score. The dial shows soil moisture at a glance. Next to it, the **plant health score** (0 to 100) blends every connected sensor against your plant's ideal ranges. Below them you can see how the device is connected, its power status, and the plant's location.

Plant profile. Tap the plant bar to tell GrowthPulse what you are growing, basil, tomatoes, succulents, ferns, and more. Every plant type has its own ideal ranges, so the colors, health score, and advice are tuned to that plant, not a generic average.

Weather rain gauge. Shows current weather at your plant's home location: temperature, conditions, rain chance, and the day's high. When rain is likely, GrowthPulse suggests skipping a watering. When intense sun and heat are coming, it warns you that soil may dry out faster. (Available on Plus and above.)

Automated watering. With a Pro plan and a pump accessory, GrowthPulse can water automatically when soil gets too dry, and it is smart enough to offer a pause when rain is on the way.

Sensors row. Every sensor with its live reading and a trend arrow. Tap any of them for a detailed chart with max, average, and minimum. A disconnected sensor is shown in gray as "not connected."

Insights. Plain-English advice generated from your readings: "Soil is very dry. Water the plant now," "Humidity is low. Consider misting," and so on. If a sensor is unplugged, Insights tells you exactly which wire to check. If a battery-powered unit is running low, the warning appears here too.

Growth journal. Add dated notes and photos to track your plant's story: repotting, pruning, first flowers. Photos appear in your shareable reports.

History

The History tab shows charts for every sensor over the last hour, day, week, or month, with your plant's ideal band shaded so you can see at a glance when conditions drifted. Max, average, and minimum are displayed for the selected window. Units follow your settings (°F or °C).

Alarms

Alarms watch your readings and alert you the moment something crosses a line you set.

- **Create an alarm** with a sensor, a direction (above or below), and a value, for example "soil moisture below 25%."
- Alarms can apply to all devices or one specific plant.
- **Suggested alarms:** GrowthPulse can create a sensible starter set tuned to your plant profile with one tap.
- When an alarm trips, a notification banner appears in the app, and with email or SMS alerts enabled in Settings, GrowthPulse contacts you even when the app is closed.

Devices

The Devices tab lists every plant on your account.

Groups. Have more than a couple of plants? Open any device's edit screen and give it a group, "Greenhouse," "Backyard," "Office." The Devices tab then organizes your plants under those headings.

Power status. Each card shows how the unit is powered: **AC power** for plugged-in units, or a battery percentage for battery-powered units, with a charging indicator. Low battery turns the badge amber, then red, and the app warns you before the unit goes offline.

Editing a device. Tap the pencil on any card to rename the plant, set or change its home location, change its group, or switch how it connects (Wi-Fi or LoRaWAN gateway).

Removing a device. In the edit screen's danger zone, "Remove from my account" takes the device off your account and deletes its history, journal photos, and alarms. The physical unit keeps working and can be paired again any time.

Selling or gifting a unit: Factory reset. Also in the danger zone. This permanently deletes everything tied to the plant from your account, and if the unit is online, it remotely wipes its own Wi-Fi and reboots into setup mode, ready for its new owner to unbox a fresh experience. Read the confirmation carefully: this cannot be undone, and if you ever reconnect the unit you will be starting from scratch. (If the unit is offline at the time, the new owner can simply hold PRG for 3 seconds.)

LoRaWAN and the Farm Kit

Wi-Fi works wonderfully indoors and around the house. For fields, large gardens, and far greenhouses, GrowthPulse offers **LoRaWAN**, long-range, low-power radio that reaches miles.

- The **Farm Kit** includes a gateway, the one piece that connects to the internet (plug it into your router). Sensor nodes then join through the gateway automatically, no Wi-Fi setup per node at all.
- Add the gateway in the app: Devices tab, **Add a LoRaWAN gateway**, enter the code on its label.
- Buying more nodes later? Power them on within range and pair with the code on their screen, the gateway part is already done.
- LoRaWAN nodes report every few minutes rather than every few seconds. That slower pace is what makes batteries last for months and signals reach for miles, perfect for plants, which do not change by the second.

Settings

- **Plan:** see your current plan and upgrade.
- **Temperature units:** °F or °C, applied everywhere instantly.
- **Theme:** light, dark, or automatic with your device.
- **Refresh rate:** how often the app updates readings, from every second to every 5 minutes.
Second-by-second rates apply to Wi-Fi devices; LoRaWAN nodes check in every few minutes by design.
Remember: on battery-powered units, faster updates mean shorter battery life.
- **Notifications:** enable email and SMS alerts and set the addresses they go to.
- **Account:** your name, grower type, email, and sign out.

Sharing a Plant Report

Tap the **document icon** in the top bar to generate a polished report for the selected plant: current readings, recent minimums, averages and maximums, the health score, weather at the plant's home, and the latest journal entries with photos. Your browser's print dialog opens automatically, choose **Save as PDF** to share it, or print it on paper. Great for plant sitters, garden clubs, agronomy classes, or just showing off.

Install the App on Your iPhone

1. Open **growthpulsecloud.com** in Safari.
2. Tap the **Share** button, then **Add to Home Screen**.
3. You get a GrowthPulse icon on your home screen, and the app opens full screen, just like an app from the App Store. Android works the same way through Chrome's "Add to home screen."

Troubleshooting

Problem	What to do
The setup page never opens after joining GrowthPulse-XXXXXX	Open a browser and go to 192.168.4.1. Also try turning off cellular data during setup.
My phone says the GrowthPulse network has no internet	That is normal during setup. Choose "Keep trying Wi-Fi" or "Use this network anyway."
The unit will not connect to my Wi-Fi	Make sure you picked the 2.4 GHz network and re-entered the password carefully. Hold PRG 3 seconds to start over.
The screen shows two dots instead of ON	The unit is still connecting. If it stays that way, your Wi-Fi may be out of range or the password changed. Hold PRG 3 seconds to redo setup.
A sensor shows "not connected"	That probe is not plugged in or its cable is loose. The Insights card names the exact sensor. Reseat the cable.
The pairing code is rejected	Codes only contain letters and numbers, check for typos (the code is on the unit's screen).
Readings look stuck	Check that the unit's screen shows ON. Pull the power and plug it back in, it reconnects by itself.
I moved or changed my router	Hold PRG 3 seconds and run Wi-Fi setup again. Nothing else changes.
The report does not open	Your browser blocked the pop-up. Allow pop-ups for growthpulsecloud.com once.

FAQ

Does GrowthPulse work without internet? The unit needs internet (through your Wi-Fi or a Farm Kit gateway) to send readings to your app. If the connection drops, it reconnects automatically when service returns.

Can multiple people watch the same plants? Each account has its own plants. Family-sharing is on our roadmap.

Does the app track my location? No. Weather uses the home location you typed for each plant. If you skip it, the app falls back to a general location estimate from your browser, which you can decline.

How accurate are the sensors? Air temperature is typically within $\pm 0.9^{\circ}\text{F}$, humidity within $\pm 2\text{-}5\%$, and soil readings are tuned for trend accuracy, watching change over time, which is what plant care decisions actually need.

Can I move a plant's probe to a different pot? Yes, just move it. Consider renaming the plant and updating its plant profile so the advice stays accurate.

What happens to my data if I remove a device? Its history, journal, photos, and alarms are deleted from your account immediately.

Specifications

Item	Specification
Power	5V USB-C, under 2.5W typical
Wi-Fi	2.4 GHz, 802.11 b/g/n
Long range option	LoRaWAN, US 915 MHz, via Farm Kit gateway
Display	Built-in OLED, shows setup help, pairing code, status
Air sensor	Temperature and relative humidity
Soil temperature	Waterproof probe
Soil moisture	Capacitive (corrosion-resistant, no exposed metal contacts)
Reporting rate	Every few seconds on Wi-Fi, every few minutes on LoRaWAN
App	growthpulsecloud.com, installable on iPhone and Android home screens

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